

# Solution to Ramanujan Challenge Problem 2.5: Efficient Rational Approximation of Catalan's Constant $G$

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## Abstract

We prove, for every one of the three terminal columns, that the matrix quotient in Problem 2.5 tends to Catalan's constant. The proof is exact. A Meijer- $G$  trajectory is related to the printed matrix by an explicit inverse-transpose identity. A positive fourfold integral identifies the unique recessive solution, and an exact Barnes-moment calculation evaluates its initial vector. We also give the exact inverse Delannoy transform and the requested finite integral formula for its coefficient error. No finite-prefix guess, decimal recognition, or non-effective sign argument is used.

## 1 The matrix and the assertion

Let  $M(n) = (m_{ij}(n))_{1 \leq i, j \leq 3}$  be the matrix in the question. For reference, its entries are

$$\begin{aligned} m_{11} &= (-2n - 5)(n + 3)^2(136n^4 + 1424n^3 + 5548n^2 + 9551n + 6141), \\ m_{12} &= 384n^6 + 6384n^5 + 44168n^4 + 162698n^3 + 336377n^2 + 369933n + 169011, \\ m_{13} &= -480n^4 - 4980n^3 - 19210n^2 - 32690n - 20730, \\ m_{21} &= (n + 2)^2(n + 3)^2(4n + 10)(48n^3 + 386n^2 + 1017n + 879), \\ m_{22} &= (n + 2)^2(-272n^5 - 3848n^4 - 21732n^3 - 61184n^2 - 85761n - 47808), \\ m_{23} &= (n + 2)^2(320n^3 + 2540n^2 + 6610n + 5640), \\ m_{31} &= (-4n - 10)(n + 2)^2(n + 3)^2(32n^4 + 302n^3 + 1037n^2 + 1530n + 813), \\ m_{32} &= (n + 2)^2(192n^6 + 2984n^5 + 19116n^4 + 64452n^3 + 120256n^2 + 117279n + 46476), \\ m_{33} &= (n + 2)^2(-16n^5 - 408n^4 - 2912n^3 - 8884n^2 - 12254n - 6240). \end{aligned}$$

Put

$$\mathcal{M}_N = M(0)M(1) \cdots M(N - 1), \quad A = \begin{pmatrix} 30921 & -32972 & 8240 \\ 33750 & -36000 & 9000 \end{pmatrix},$$

where the empty product is the identity, and define

$$A\mathcal{M}_N = \begin{pmatrix} P_{N,1} & P_{N,2} & P_{N,3} \\ Q_{N,1} & Q_{N,2} & Q_{N,3} \end{pmatrix}.$$

Finally,

$$G = \sum_{r=0}^{\infty} \frac{(-1)^r}{(2r + 1)^2}.$$

**Theorem 1** (Main assertion). *For  $j = 1, 2, 3$ ,*

$$\lim_{N \rightarrow \infty} \frac{P_{N,j}}{Q_{N,j}} = G.$$

The proof is completed in Theorem 19.

**Lemma 2** (Catalan moment). *The improper integral is absolutely convergent and*

$$G = \int_0^1 \frac{-\log t}{1+t^2} dt, \quad \int_0^1 (-\log t) dt = 1.$$

*Proof.* The second identity follows by integration by parts. For  $0 \leq t < 1$ ,

$$\frac{1}{1+t^2} = \sum_{r \geq 0} (-1)^r t^{2r}.$$

The partial sums have absolute value at most 2, while  $-\log t$  is integrable on  $(0, 1)$ . Dominated convergence and  $\int_0^1 t^{2r} (-\log t) dt = (2r+1)^{-2}$  give the first identity.  $\square$

The following exact moment identity will also be used in the Delannoy appendix. It is not used to infer decay merely from a finite alternating sum.

**Lemma 3** (Exact moment lift). *Let  $\mathbf{p}_0$  and  $\mathbf{q}_0$  be the two rows of  $A$ , set*

$$\mathbf{R}_0(X) = \mathbf{q}_0 - (1+X)\mathbf{p}_0, \quad R_{N,j}(X) = (\mathbf{R}_0(X)\mathcal{M}_N)_j.$$

*Then  $R_{N,j}(X) = Q_{N,j} - (1+X)P_{N,j}$  and*

$$Q_{N,j} = \int_0^1 (10-18x)R_{N,j}(x) dx, \tag{1}$$

$$P_{N,j} = \int_0^1 (6-12x)R_{N,j}(x) dx, \tag{2}$$

$$GQ_{N,j} - P_{N,j} = \int_0^1 \frac{-\log t}{1+t^2} R_{N,j}(t^2) dt. \tag{3}$$

*Proof.* The displayed affine formula for  $R_{N,j}$  follows directly from the definition. If  $R(X) = a+bX$ , then the four elementary moments

$$\int_0^1 (10-18x) dx = 1, \quad \int_0^1 x(10-18x) dx = -1, \quad \int_0^1 (6-12x) dx = 0, \quad \int_0^1 x(6-12x) dx = -1$$

give (1) and (2). Lemma 2 gives

$$\int_0^1 \frac{-\log t}{1+t^2} (Q - (1+t^2)P) dt = GQ - P,$$

which is (3). The logarithmic endpoint is integrable, so these are ordinary improper Lebesgue integrals.  $\square$

## 2 Inverse-transpose selection

We record the standard form of the recessive-solution principle needed below. It is a consequence of the Poincaré–Perron theorem in its higher-order (discrete Levinson) form; see [1, Sec. 3.9] and [2].

**Proposition 4** (Recessive recognition and Miller selection). *Suppose a row companion system  $X_{n+1} = X_n C(n)$  satisfies  $C(n) = C_\infty + O(n^{-1})$ , where  $C_\infty$  is invertible and diagonalizable. Suppose  $C_\infty$  has a simple eigenvalue  $\mu$  such that*

$$0 < |\mu| < \min\{|\lambda| : \lambda \in \text{spec}(C_\infty), \lambda \neq \mu\}.$$

*If a nonzero scalar solution  $F_n$  has  $\limsup |F_n|^{1/n} \leq |\mu|$ , it spans the unique recessive solution line. Writing*

$$\Phi_C(N) = C(0) \cdots C(N-1), \quad \mathbf{F}_0 = (F_0, \dots, F_{r-1})^T,$$

*one has*

$$[\Phi_C(N)^{-T} w_N] \longrightarrow [\mathbf{F}_0]$$

*for every terminal vector  $w_N$  whose coefficient in the corresponding dominant dual direction is nonzero.*

*Moreover, if  $T(n)U(n+1) = U(n)C(n)$  with rational invertible  $U(n)$ , then*

$$[\Phi_T(N)^{-T} e_j] \longrightarrow [U(0)^{-T} \mathbf{F}_0] \tag{4}$$

*whenever  $U(N)^T e_j$  has nonzero recessive-dual connection.*

*Proof.* The spectral gap splits the asymptotic solution space into the one-dimensional  $\mu$ -space and its complementary block. Discrete Levinson asymptotics show that every solution outside the first space has strictly larger exponential rate. This proves the recognition statement. Inverting and transposing a fundamental product replaces each exponential factor  $\lambda^N$  by  $\lambda^{-N}$ ; hence the recessive primal line becomes the unique dominant dual line. This proves the projective assertion.

Finally, multiplication of the coboundary identities gives

$$\Phi_T(N) = U(0)\Phi_C(N)U(N)^{-1}, \quad \Phi_T(N)^{-T} = U(0)^{-T}\Phi_C(N)^{-T}U(N)^T.$$

Rational gauges have at most polynomial growth along the positive integers and therefore do not alter an exponentially isolated line. This gives (4).  $\square$

## 3 An exact Meijer– $G$ trajectory

We use the Barnes convention of [3, Sec. 16.17],

$$G_{p,q}^{m,r} \left( z \left| \begin{matrix} a_1, \dots, a_p \\ b_1, \dots, b_q \end{matrix} \right. \right) = \frac{1}{2\pi i} \int_L \frac{\prod_{j=1}^m \Gamma(b_j - s) \prod_{j=1}^r \Gamma(1 - a_j + s)}{\prod_{j=m+1}^q \Gamma(1 - b_j + s) \prod_{j=r+1}^p \Gamma(a_j - s)} z^s ds.$$

For  $n \geq 0$  define

$$F_n(z) = G_{4,4}^{4,2} \left( z \left| \begin{matrix} -1-n, -1-n, n+\frac{5}{2}, n+3 \\ \frac{3}{2}, \frac{1}{2}, 1, 0 \end{matrix} \right. \right), \quad F_n = F_n(1), \tag{5}$$

and set  $\Theta = z d/dz$  and

$$Y_n = (F_n, (\Theta F_n)(1), (\Theta^2 F_n)(1)).$$

**Lemma 5** (Differential reduction). *At  $z = 1$ ,*

$$(\Theta^3 F_n)(1) = -\frac{(n+2)^3(2n+3)}{7}F_n + \frac{2n^2+8n+5}{14}(\Theta F_n)(1) + \frac{8n^2+30n+39}{14}(\Theta^2 F_n)(1). \quad (6)$$

*Proof.* The Meijer- $G$  differential equation for (5) is

$$[z(\Theta + n + 2)^2(\Theta - n - \frac{3}{2})(\Theta - n - 2) - (\Theta - \frac{3}{2})(\Theta - \frac{1}{2})(\Theta - 1)\Theta] F_n(z) = 0.$$

At  $z = 1$  the fourth-order terms cancel. Expansion and division by  $7/2$  gives (6).  $\square$

Define  $T(n) = (t_{ij}(n))$  by

$$\begin{pmatrix} \frac{4(n+2)(17n^3+111n^2+240n+171)}{(n+1)(n+3)(2n+3)(2n+5)} & \frac{(n+2)(24n^2+101n+102)}{(n+1)(2n+3)} & \frac{(n+2)(2n+5)(16n^2+81n+90)}{2(n+1)(2n+3)} \\ \frac{96n^4+780n^3+2384n^2+3273n+1723}{(n+1)(n+2)(n+3)(2n+3)(2n+5)} & \frac{68n^3+398n^2+778n+523}{2(n+1)(n+2)(2n+3)} & \frac{96n^4+884n^3+2970n^2+4360n+2403}{4(n+1)(n+2)(2n+3)} \\ \frac{-5(24n^2+117n+143)}{(n+1)(n+2)(n+3)(2n+3)(2n+5)} & \frac{-5(16n+41)}{2(n+1)(n+2)(2n+3)} & \frac{8n^3-44n^2-478n-801}{4(n+1)(n+2)(2n+3)} \end{pmatrix}.$$

**Lemma 6** (Exact contiguous trajectory). *For every integer  $n \geq 0$ ,*

$$Y_{n+1} = Y_n T(n).$$

*Proof.* In the Barnes integrand for  $F_n$ , the  $n$ -dependent factor is

$$\frac{\Gamma(n+2+s)^2}{\Gamma(n+\frac{5}{2}-s)\Gamma(n+3-s)}.$$

Its quotient after  $n \mapsto n+1$  is

$$\frac{(n+2+s)^2}{(n+\frac{5}{2}-s)(n+3-s)}.$$

Apply this identity also after multiplication by  $s$  and  $s^2$ . Gamma recurrence and the cubic reduction (6) reduce the three resulting Barnes integrals to the basis  $1, s, s^2$ ; the three coefficient columns are exactly the displayed columns of  $T(n)$ . All operations are identities in  $\mathbb{Q}(n, s)$  after the gamma factors are cancelled. They are first valid on a Barnes contour for  $0 < z < 1$ ; gamma decay permits contour shifting and differentiation. Dominated passage to  $z = 1$  gives the assertion. Thus this is an exact contiguous calculation, not a check of finitely many values.  $\square$

Put

$$\sigma(n) = -2(n+2)^2(n+3)^2(2n+5)(2n+7)^2. \quad (7)$$

**Lemma 7** (Inverse-transpose certificate). *For the printed matrix  $M(n)$  and the displayed  $T(n)$ ,*

$$T(n)^T M(n) = \sigma(n) I_3. \quad (8)$$

*Consequently, if  $\Phi_T(N) = T(0) \cdots T(N-1)$  and  $H_N = \prod_{m=0}^{N-1} \sigma(m)$ , then*

$$\mathcal{M}_N = H_N \Phi_T(N)^{-T}, \quad H_N = (-16)^N (2)_N^2 (3)_N^2 (\frac{5}{2})_N (\frac{7}{2})_N^2. \quad (9)$$

*Proof.* Substitution of the nine displayed rational functions and the nine polynomials defining  $M(n)$  gives (8) after clearing the nonzero denominator  $(n+1)(n+2)(n+3)(2n+3)(2n+5)$ . This is a nine-entry polynomial identity in  $\mathbb{Q}[n]$ . Multiplying it for  $0 \leq n < N$  gives the first identity in (9); factoring each linear term in (7) gives the stated Pochhammer product.  $\square$

## 4 Companion form and exact connection data

Let  $e_1 = (1, 0, 0)^T$  and define the Krylov matrix

$$U(n) = [e_1, T(n)e_1, T(n)T(n+1)e_1]. \quad (10)$$

**Lemma 8** (Krylov determinant). *Let*

$$D(n) = 3072n^7 + 61824n^6 + 531184n^5 + 2525522n^4 + 7175793n^3 + 12183785n^2 + 11446123n + 4589916.$$

*Then*

$$\det U(n) = -\frac{15D(n)}{2(n+1)(n+2)^3(n+3)^2(n+4)(2n+3)^2(2n+5)^2(2n+7)}.$$

*In particular,  $U(n)$  is invertible for every integer  $n \geq 0$ .*

*Proof.* Expand the determinant using (10) and factor. Every coefficient of  $D$  is positive and all denominator factors are positive for  $n \geq 0$ .  $\square$

**Lemma 9** (Companion coboundary). *The matrix*

$$C(n) = U(n)^{-1}T(n)U(n+1)$$

*is a row companion matrix,  $T(n)U(n+1) = U(n)C(n)$ , and*

$$C(n) = C_\infty + O(n^{-1}), \quad C_\infty = \begin{pmatrix} 0 & 0 & 1 \\ 1 & 0 & -35 \\ 0 & 1 & 35 \end{pmatrix}.$$

*Moreover*

$$Y_n U(n) = (F_n, F_{n+1}, F_{n+2}). \quad (11)$$

*Proof.* The first two columns of the coboundary identity follow from the definition of  $U(n)$ ; the last column defines the scalar recurrence, so  $C(n)$  is companion. Exact rational simplification gives the displayed limit. Equation (11) follows from Lemma 6.  $\square$

**Lemma 10** (Spectral gap). *The characteristic polynomial of  $C_\infty$  is*

$$(\lambda - 1)(\lambda^2 - 34\lambda + 1).$$

*Its roots are*

$$\mu = 17 - 12\sqrt{2} = (\sqrt{2} - 1)^4, \quad 1, \quad \mu^{-1} = 17 + 12\sqrt{2}.$$

*Thus  $\mu$  is simple and has strictly smallest modulus. A right eigenvector for  $\mu$  is*

$$r_\mu = (17 + 12\sqrt{2}, -18 - 12\sqrt{2}, 1)^T. \quad (12)$$

*Proof.* This is direct factorization and multiplication by  $C_\infty$ .  $\square$

**Lemma 11** (All three terminal connections are nonzero). *For each  $j = 1, 2, 3$ , the vector  $U(N)^T e_j$  has nonzero connection with the recessive dual mode in Proposition 4.*

*Proof.* Exact leading terms of the displayed rational matrix  $U(n)$  give

$$\lim_{n \rightarrow \infty} \text{diag}(1, n, n^3)U(n) = L := \begin{pmatrix} 1 & 17 & 577 \\ 0 & 24 & 816 \\ 0 & -30 & -1020 \end{pmatrix}. \quad (13)$$

The three row pairings of  $L$  with (12) are

$$96(3 - 2\sqrt{2}), \quad 96(4 - 3\sqrt{2}), \quad 120(-4 + 3\sqrt{2}).$$

None is zero: otherwise  $\sqrt{2}$  would be rational. Polynomial row scalings do not change whether the corresponding asymptotic coefficient vanishes. Hence all three terminal vectors satisfy the noncancellation hypothesis.  $\square$

Let  $\mathbf{F} = (F_0, F_1, F_2)^T$ .

**Lemma 12** (Barnes initial values). *The initial vector is*

$$\mathbf{F} = \sqrt{\pi} \begin{pmatrix} 150G - 128 \log 2 - \frac{146}{3} \\ \frac{24745}{7225281}G - \frac{14624}{886784} \log 2 - \frac{823511}{2818419551} \\ \frac{4}{32}G - \frac{3}{5} \log 2 - \frac{360}{33600} \end{pmatrix}. \quad (14)$$

*Proof.* Insert  $n = 0, 1, 2$  in the Barnes integrand in (5). Cancelling the two gamma quotients gives

$$\frac{\Gamma(\frac{1}{2} - s)\Gamma(-s)\Gamma(n + 2 + s)^2}{(\frac{3}{2} - s)_{n+1}(1 - s)_{n+2}}.$$

Residue expansion (first at  $z = r$  with  $0 < r < 1$  and then with  $r \uparrow 1$ ) reduces each of the three cases by partial fractions to

$$\sum_{k \geq 0} \frac{(-1)^k}{2k+1} = \frac{\pi}{4}, \quad \sum_{k \geq 1} \frac{(-1)^{k-1}}{k} = \log 2, \quad \sum_{k \geq 0} \frac{(-1)^k}{(2k+1)^2} = G,$$

plus finite rational tails. Collecting the coefficients gives (14). This is the exact Barnes-moment evaluation recorded independently in [1, Lemma 19]; the regulator  $r$  also justifies the contour endpoint before dominated passage to  $r = 1$ .  $\square$

**Lemma 13** (Initial coboundary). *One has*

$$U(0) = \begin{pmatrix} 1 & \frac{152}{5} & \frac{195477}{175} \\ 0 & \frac{1723}{90} & \frac{1963751}{2800} \\ 0 & -\frac{143}{18} & -\frac{165201}{560} \end{pmatrix}$$

and

$$AU(0)^{-T} = \frac{1}{382493} \begin{pmatrix} 30102216645 & -19896711516 & 525137760 \\ 32863650750 & -21712357800 & 573048000 \end{pmatrix}. \quad (15)$$

*Proof.* Substitute  $n = 0$  in (10); exact inversion and multiplication by  $A$  give (15).  $\square$

**Lemma 14** (Exact Catalan cancellation). *The logarithmic and rational terms cancel exactly:*

$$AU(0)^{-T}\mathbf{F} = 450\sqrt{\pi} \begin{pmatrix} G \\ 1 \end{pmatrix}. \quad (16)$$

*Proof.* Multiply the rational matrix (15) by (14). The two coefficients of  $\log 2$  are zero, the two rational constant terms are zero, and the remaining coefficients are respectively  $450G$  and  $450$ . This is an identity in the three-dimensional  $\mathbb{Q}$ -space spanned by  $1, \log 2, G$ ; no decimal recognition is involved.  $\square$

## 5 Exact Delannoy transform

This section repairs the finite-prefix claims in the earlier Delannoy argument. It is included because the transform and its error integral are useful exact identities. The main convergence proof below does not require a sign assertion about the transformed coefficient  $f(k)$ .

Let

$$\mathcal{B}(N, k) = 2^k \binom{2k}{k} \binom{N}{k} \binom{N+k}{k} \quad (0 \leq k \leq N).$$

**Lemma 15** (Inverse Delannoy matrix). *The inverse of the lower-triangular matrix  $\mathcal{B}$  is*

$$\beta(k, N) = \frac{(-1)^{k+N} (2N+1) \binom{k}{N}}{2^k \binom{2k}{k} (k+N+1) \binom{k+N}{N}} \quad (0 \leq N \leq k). \quad (17)$$

*Proof.* We prove  $\sum_{N=j}^k \beta(k, N) \mathcal{B}(N, j) = \delta_{kj}$ . For  $k = j$  the sole product is 1. Suppose  $m = k - j > 0$  and put  $N = j + r$ . After removing a nonzero factor independent of  $r$  and using

$$\frac{2j + 2r + 1}{k + j + r + 1} = 2 - \frac{2k + 1}{k + j + r + 1},$$

the sum becomes

$$\begin{aligned} & 2 {}_2F_1(-m, 2j+1; k+j+1; 1) \\ & - \frac{2k+1}{k+j+1} {}_2F_1(-m, 2j+1; k+j+2; 1). \end{aligned}$$

Chu–Vandermonde evaluates the two hypergeometric terms as

$$\frac{(m)_m}{(k+j+1)_m}, \quad \frac{(m+1)_m}{(k+j+2)_m}.$$

Their ratio is  $2(k+j+1)/(2k+1)$ , so the displayed difference is zero. This proves the inverse formula for every index, rather than checking a finite square.  $\square$

Define the scalar normalization

$$H_N = (-16)^N (2)_N^2 (3)_N^2 \left(\frac{5}{2}\right)_N \left(\frac{7}{2}\right)_N^2$$

and, for the first column only in this section,

$$\hat{P}_N = P_{N,1}/H_N, \quad \hat{Q}_N = Q_{N,1}/H_N.$$

There are unique rational sequences  $f, g$  such that

$$\hat{Q}_N = \sum_{k=0}^N f(k) \mathcal{B}(N, k), \quad \hat{P}_N = \sum_{k=0}^N g(k) \mathcal{B}(N, k), \quad (18)$$

namely

$$f(k) = \sum_{N=0}^k \beta(k, N) \hat{Q}_N, \quad g(k) = \sum_{N=0}^k \beta(k, N) \hat{P}_N. \quad (19)$$

**Theorem 16** (Holonomic  $k$ -recurrences). *The sequences  $f(k)$  and  $g(k)$  in (19) are  $P$ -recursive. This conclusion holds for all  $k$  as a consequence of holonomic closure applied to the exact finite sums; it is not based on a guessed operator or verification of a finite prefix.*

*Proof.* By (9),  $\hat{P}_N$  and  $\hat{Q}_N$  are coordinates of a fixed row multiplied by the three-dimensional rational first-order system  $\Phi_T(N)^{-T}$ . Four consecutive coordinate functionals are linearly dependent over  $\mathbb{Q}(N)$ ; clearing denominators gives a polynomial-coefficient scalar recurrence. Thus both sequences are holonomic in  $N$ .

The kernel  $\beta(k, N)$  is proper hypergeometric: both of its shift quotients are rational functions of  $(k, N)$ . Zeilberger's creative telescoping theorem [4], equivalently closure of holonomic functions under definite summation, applied to the finite range  $0 \leq N \leq k$  therefore produces polynomial-coefficient recurrences for the two sums (19). The argument proves existence and annihilation identically; no order, degree, or minimality inferred by guessing is asserted.  $\square$

**Theorem 17** (Effective positivity and recessiveness). *For every  $n \geq 0$ ,  $F_n > 0$ , and*

$$\lim_{n \rightarrow \infty} F_n^{1/n} = 17 - 12\sqrt{2} = \mu. \quad (20)$$

*Consequently  $(F_n)$  is the unique recessive solution of the companion recurrence in Lemma 9.*

*Proof.* Two beta integrals, two Laplace integrals, and the duplication formula transform the Barnes integral exactly into

$$F_n = \frac{\sqrt{\pi}}{\Gamma(n+1)\Gamma(n+2)} \int_{(0,1)^2 \times (0,\infty)^2} x^{1/2}(1-x)^n(1-y)^{n+1}u^{n+1}v^{n+1} \times \exp\left(-u-v-2\sqrt{\frac{uv}{xy}}\right) du dv dx dy. \quad (21)$$

Every factor is positive, proving  $F_n > 0$  directly for every  $n$ .

For the exact root rate put  $u = na$ ,  $v = nb$  and write

$$F_n = \frac{\sqrt{\pi} n^{2n+4}}{\Gamma(n+1)\Gamma(n+2)} J_n, \quad J_n = \int_D A e^{n\Phi} dx dy da db,$$

where  $D = (0,1)^2 \times (0,\infty)^2$ ,

$$A = x^{1/2}(1-y)ab$$

and

$$\Phi = \log(1-x) + \log(1-y) + \log a + \log b - a - b - 2\sqrt{ab/(xy)}.$$

Let  $p = \sqrt{xy}$  and  $r = \sqrt{ab}$ . Arithmetic-geometric mean gives

$$\Phi \leq 2\log(1-p) + 2\log r - 2r(1+p^{-1}),$$

with equality only when  $x = y$  and  $a = b$ . Maximizing first over  $r$  and then over  $p$  yields

$$\Phi + 2 \leq 2\log \frac{p(1-p)}{1+p} \leq 4\log(\sqrt{2}-1) =: \Phi_* + 2.$$

The unique maximizer is

$$x = y = \sqrt{2} - 1, \quad a = b = \frac{\sqrt{2} - 1}{\sqrt{2}}.$$

Moreover

$$Ae^\Phi \leq x^{1/2}(1-x)(1-y)^2 a^2 b^2 e^{-a-b},$$

and the right-hand side is integrable on  $D$ . Hence  $J_n \leq e^{(n-1)\Phi_*} \int_D Ae^\Phi$ . Conversely, on a fixed small neighborhood of the unique maximizer,  $A$  is bounded below by a positive constant and  $\Phi \geq \Phi_* - \varepsilon$ . It follows that  $J_n^{1/n} \rightarrow e^{\Phi_*}$ ; this is the elementary Laplace principle with explicit maximizer, not a finite numerical check. Stirling's formula gives

$$\left( \frac{\sqrt{\pi} n^{2n+4}}{\Gamma(n+1)\Gamma(n+2)} \right)^{1/n} \rightarrow e^2.$$

Therefore

$$\lim F_n^{1/n} = e^{\Phi_*+2} = (\sqrt{2}-1)^4 = 17 - 12\sqrt{2}.$$

Lemma 10 and Proposition 4 now identify  $(F_n)$  as the unique recessive solution.  $\square$

**Theorem 18** (Exact integral for the Delannoy coefficient error). *Put  $h(k) = g(k) - Gf(k)$  and*

$$\Psi_k(t) = \sum_{N=0}^k \beta(k, N) \frac{R_{N,1}(t^2)}{H_N}. \quad (22)$$

*Then, for every  $k \geq 0$ ,*

$$h(k) = - \int_0^1 \frac{-\log t}{1+t^2} \Psi_k(t) dt. \quad (23)$$

*The sum in (22) is finite and has rational coefficients; in fact*

$$\Psi_k(t) = f(k) - (1+t^2)g(k). \quad (24)$$

*Proof.* Divide (3) by  $H_N$ , multiply by  $\beta(k, N)$ , and sum over  $0 \leq N \leq k$ . The sum has only  $k+1$  terms, so its interchange with the absolutely convergent integral is automatic. Equations (19) give (23). Applying the same inverse sum to  $R_{N,1}(t^2)/H_N = \widehat{Q}_N - (1+t^2)\widehat{P}_N$  gives (24).  $\square$

*Remark 1.* Equation (24) shows why the integral identity alone must not be advertised as a decay estimate: without an independent bound it is an exact rewriting of  $h(k)$ . The earlier inference from finite cancellation to an all- $k$   $O(8^{-k})$  bound is therefore deleted. The positive integral (21) supplies the effective all-index argument actually needed for the proof.

*Remark 2* (Other audit corrections). If the unused raw scalar recurrence is indexed as

$$\sum_{j=0}^3 c_j(n) y_{n+j} = 0,$$

the degree pattern is

$$(28, 21, 14, 7), \quad \deg c_j = 7(4-j),$$

not  $\deg c_j = 7(3-j)$ . The former modular paragraph has also been removed: the eta product  $\eta^2(2\tau)\eta^2(4\tau)$  has formal weight 2, and no modular claim is required here. All endpoint passages used above are made by dominated convergence or by the positive Laplace integral.

## 6 Limit identification and all three columns

**Theorem 19** (Hatted sequences and the three-column bridge). *For every  $N \geq 0$  and  $j = 1, 2, 3$ , define*

$$\hat{P}_{N,j} = \frac{P_{N,j}}{H_N}, \quad \hat{Q}_{N,j} = \frac{Q_{N,j}}{H_N}, \quad H_N = (-16)^N (2)_N^2 (3)_N^2 \left(\frac{5}{2}\right)_N \left(\frac{7}{2}\right)_N^2. \quad (25)$$

*Then*

$$\frac{P_{N,j}}{Q_{N,j}} = \frac{\hat{P}_{N,j}}{\hat{Q}_{N,j}} \longrightarrow G \quad (j = 1, 2, 3).$$

*Proof.* By (9), for each standard column  $e_j$ ,

$$\begin{pmatrix} \hat{P}_{N,j} \\ \hat{Q}_{N,j} \end{pmatrix} = A\Phi_T(N)^{-T} e_j. \quad (26)$$

Theorem 17 identifies  $\mathbf{F}$  as the recessive initial vector. Lemma 11 verifies the nonzero terminal coefficient separately for each of  $j = 1, 2, 3$ . Proposition 4 and Lemma 14 therefore give

$$[A\Phi_T(N)^{-T} e_j] \longrightarrow [AU(0)^{-T} \mathbf{F}] = \left[ 450\sqrt{\pi} \begin{pmatrix} G \\ 1 \end{pmatrix} \right] = \left[ \begin{pmatrix} G \\ 1 \end{pmatrix} \right].$$

The second coordinate of a normalized representative tends to the nonzero number 1, so  $\hat{Q}_{N,j}$  (and hence  $Q_{N,j}$ ) is nonzero for all sufficiently large  $N$ , and  $\hat{P}_{N,j}/\hat{Q}_{N,j} \rightarrow G$ . Finally  $H_N \neq 0$ , so the equality of the hatted and original ratios in (25) proves all three assertions of Theorem 1.  $\square$

## References

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